Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100504 | *Subject: Microprocessor* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) Which of the following is the brain of a microcomputer system?
(i) RAM (iii) ROM
(ii) Microprocessor (iv) ALU

(b) Which company developed the 8051 microcontroller?
(i) Intel (iii) IBM
(ii) AMD (iv) Motorola

(c) The Stack Pointer points to
(i) Program memory (iii) Data memory
(ii) Top of stack (iv) I/O port

(d) Which port has dual functionality in 8051?
(i) Port 0 (iii) Port 1
(ii) Port 2 (iv) Port 3

(e) Indexed addressing mode is mainly used for
(i) Arrays (iii) Stack
(ii) Arithmetic (iv) Logic

(f) The 8051 has how many I/O ports?
(i) 1 (iii) 2
(ii) 4 (iv) 5

(g) Which instruction decrements and jumps if not zero?
(i) DJNZ (iii) CJNE
(ii) AJMP (iv) NOP

(h) I2C uses how many lines?
(i) 1 (iii) 2
(ii) 4 (iv) 8

(i) Interfacing a DC motor generally requires
(i) Driver circuit (iii) Keyboard
(ii) Counter (iv) Crystal oscillator

(j) Which protocol is commonly used for sensor interfacing?
(i) I2C (iii) HTTP
(ii) FTP (iv) SMTP

Q.2 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the architecture of an 8-bit microprocessor with a neat block diagram.

(b) Discuss the advantages and disadvantages of embedded systems.

Q.3 **Solve both questions :** **[7 x 2 = 14]**

(a) Draw and explain the internal block diagram of the 8051 microcontroller.

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(b) Describe the role of Program Counter and timing circuits in 8051.

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the Special Function Registers used in 8051.

(b) Discuss the data transfer and arithmetic instructions of the 8051.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Write a program to find the largest number in an array using 8051 assembly language.

(b) Explain Different Addressing Modes of 8051 Microcontroller.

Q.6 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain subroutine instructions and stack operations in 8051.

(b) Explain memory interfacing techniques in the 8051 microcontroller.

Q.7 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the interfacing of ADC with the 8051 microcontroller.

(b) Discuss the RS232 communication interface with neat diagrams.

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the interfacing of wireless communication protocols in embedded systems.

(b) Discuss LCD interfacing with neat diagrams.

Q.9 **Write short notes on any four of the following:** **[3½ x 4 = 14]**

(a) Compare microprocessor and microcontroller with suitable examples

(b) Explain the address bus, data bus, and control bus in the 8051 architecture

(c) Explain the concept of wait states in memory interfacing

(d) Discuss Zigbee protocol and its advantages

(e) Explain the working and applications of stepper motors

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